

Data Preprocessing Methodology

1 Introduction

To prepare the raw textual data for training, a preprocessing pipeline was implemented. The objective of this pipeline is to standardize the textual features and eliminate noise. The preprocessing workflow used the `pandas` library for dataframe manipulation, the `re` module for regular expressions, and the Natural Language Toolkit (NLTK) for lexical processing.

2 Preprocessing Methodology

2.1 Missing Value Handling

Missing values are handled at the individual record level during text cleaning. When a text field is found to be null or missing (e.g., `NaN` or `None`) the cleaning function returns an empty string for that record.

2.2 Case Normalization

All textual data is converted to lowercase. This standardizes the vocabulary by ensuring that capitalized and uncapitalized instances of the same word (e.g., “Study” and “study”) are treated as identical tokens.

2.3 Greek Character Transliteration

Academic and scientific texts frequently use Greek characters, to preserve this information while standardizing the character set, all Greek letters are mapped into their corresponding English phonetic spellings (e.g., ‘ α ’ becomes ‘alpha’).

2.4 Special Character and Punctuation Removal

A regular expression filter is applied to remove punctuation, symbols, and non-alphanumeric noise. The algorithm retains only lowercase alphabetical characters (`a-z`), numerical digits (`0-9`), spaces, and hyphens (`-`). All other characters are substituted with a space.

2.5 Whitespace Normalization

Artifacts from the previous cleaning steps often result in irregular spacing. Regular expressions are utilized to compress multiple consecutive whitespace characters into a single space. Additionally, leading and trailing whitespaces are stripped from the text.

2.6 Tokenization and Stop Word Removal

The normalized text is tokenized into individual words by splitting the string on whitespace. To reduce vocabulary dimensionality and discard uninformative terms, a comprehensive stop word removal process is executed. The exclusion filter combines:

- The English stop words corpus provided by the NLTK library.
- A custom, domain-specific dictionary of academic stop words. This includes terms that provide structural rather than semantic value to scientific papers, specifically: *doi, et, al, figure, table, data, used, show, found, based, study, and result.*

Tokens matching any term in this combined exclusion list are discarded. The surviving semantic tokens are then re-joined into a single, continuous string.